

Week 2

```
In [4]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Working with pandas

```
In [5]: df = pd.read_csv('data/obesity.csv')
```

```
In [6]: df.info()

<class 'pandas.DataFrame'>
RangeIndex: 2111 entries, 0 to 2110
Data columns (total 17 columns):
 #   Column              Non-Null Count  Dtype
---  --
 0   Gender              2111 non-null  str
 1   Age                 2111 non-null  float64
 2   Height              2111 non-null  float64
 3   Weight              2111 non-null  float64
 4   family_history_with_overweight  2111 non-null  str
 5   FAVC                2111 non-null  str
 6   FCVC                2111 non-null  float64
 7   NCP                 2111 non-null  float64
 8   CAEC                2111 non-null  str
 9   SMOKE               2111 non-null  str
10  CH2O                2111 non-null  float64
11  SCC                 2111 non-null  str
12  FAF                 2111 non-null  float64
13  TUE                 2111 non-null  float64
14  CALC                2111 non-null  str
15  MTRANS              2111 non-null  str
16  NObesyesdad         2111 non-null  str
dtypes: float64(8), str(9)
memory usage: 288.5 KB
```

```
In [7]: df.describe()
```

	Age	Height	Weight	FCVC	NCP	CH2O	FAF	TUE
count	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000	2111.000000
mean	24.312600	1.701677	86.586058	2.419043	2.685628	2.008011	1.010298	0.657866
std	6.345968	0.093305	26.191172	0.533927	0.778039	0.612953	0.850592	0.608927
min	14.000000	1.450000	39.000000	1.000000	1.000000	1.000000	0.000000	0.000000
25%	19.947192	1.630000	65.473343	2.000000	2.658738	1.584812	0.124505	0.000000
50%	22.777890	1.700499	83.000000	2.385502	3.000000	2.000000	1.000000	0.625350
75%	26.000000	1.768464	107.430682	3.000000	3.000000	2.477420	1.666678	1.000000
max	61.000000	1.980000	173.000000	3.000000	4.000000	3.000000	3.000000	2.000000

```
In [8]: df.head()
```

Out[8]:	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	MTRANS	NObesyesdad
0	Female	21.0	1.62	64.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	0.0	1.0	no	Public_Transportation	Normal_Weight
1	Female	21.0	1.52	56.0	yes	no	3.0	3.0	Sometimes	yes	3.0	yes	3.0	0.0	Sometimes	Public_Transportation	Normal_Weight
2	Male	23.0	1.80	77.0	yes	no	2.0	3.0	Sometimes	no	2.0	no	2.0	1.0	Frequently	Public_Transportation	Normal_Weight
3	Male	27.0	1.80	87.0	no	no	3.0	3.0	Sometimes	no	2.0	no	2.0	0.0	Frequently	Walking	Overweight_Level_I
4	Male	22.0	1.78	89.8	no	no	2.0	1.0	Sometimes	no	2.0	no	0.0	0.0	Sometimes	Public_Transportation	Overweight_Level_II

```
In [9]: df.tail()
```

Out(9):	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	MTRANS	NObesyesdad	
2106	Female	20.976842	1.710730	131.408528		yes	yes	3.0	3.0	Sometimes	no	1.728139	no	1.676269	0.906247	Sometimes	Public_Transportation	Obesity_Type_III
2107	Female	21.982942	1.748584	133.742943		yes	yes	3.0	3.0	Sometimes	no	2.005130	no	1.341390	0.599270	Sometimes	Public_Transportation	Obesity_Type_III
2108	Female	22.524036	1.752206	133.689352		yes	yes	3.0	3.0	Sometimes	no	2.054193	no	1.414209	0.646288	Sometimes	Public_Transportation	Obesity_Type_III
2109	Female	24.361936	1.739450	133.346641		yes	yes	3.0	3.0	Sometimes	no	2.852339	no	1.139107	0.586035	Sometimes	Public_Transportation	Obesity_Type_III
2110	Female	23.664709	1.738836	133.472641		yes	yes	3.0	3.0	Sometimes	no	2.863513	no	1.026452	0.714137	Sometimes	Public_Transportation	Obesity_Type_III

```
In [14]: df_size = 5
df.head(df_size)
```

```
Out[14]:
```

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	MTRANS	NObeyesdad	
0	Female	21.0	1.62	64.0		yes	no	2.0	3.0	Sometimes	no	2.0	no	0.0	1.0	no	Public_Transportation	Normal_Weight
1	Female	21.0	1.52	56.0		yes	no	3.0	3.0	Sometimes	yes	3.0	yes	3.0	0.0	Sometimes	Public_Transportation	Normal_Weight
2	Male	23.0	1.80	77.0		yes	no	2.0	3.0	Sometimes	no	2.0	no	2.0	1.0	Frequently	Public_Transportation	Normal_Weight
3	Male	27.0	1.80	87.0		no	no	3.0	3.0	Sometimes	no	2.0	no	2.0	0.0	Frequently	Walking	Overweight_Level_I
4	Male	22.0	1.78	89.8		no	no	2.0	1.0	Sometimes	no	2.0	no	0.0	0.0	Sometimes	Public_Transportation	Overweight_Level_II

```
In [15]: df.tail(df_size)
```

```
Out[15]:
```

	Gender	Age	Height	Weight	family_history_with_overweight	FAVC	FCVC	NCP	CAEC	SMOKE	CH2O	SCC	FAF	TUE	CALC	MTRANS	NObeyesdad	
2106	Female	20.976842	1.710730	131.408528		yes	yes	3.0	3.0	Sometimes	no	1.728139	no	1.676269	0.906247	Sometimes	Public_Transportation	Obesity_Type_III
2107	Female	21.982942	1.748584	133.742943		yes	yes	3.0	3.0	Sometimes	no	2.005130	no	1.341390	0.599270	Sometimes	Public_Transportation	Obesity_Type_III
2108	Female	22.524036	1.752206	133.689352		yes	yes	3.0	3.0	Sometimes	no	2.054193	no	1.414209	0.646288	Sometimes	Public_Transportation	Obesity_Type_III
2109	Female	24.361936	1.739450	133.346641		yes	yes	3.0	3.0	Sometimes	no	2.852339	no	1.139107	0.586035	Sometimes	Public_Transportation	Obesity_Type_III
2110	Female	23.664709	1.738836	133.472641		yes	yes	3.0	3.0	Sometimes	no	2.863513	no	1.026452	0.714137	Sometimes	Public_Transportation	Obesity_Type_III

```
In [13]: print(df["Age"].mean()) # average value, sum of all values / number of times
print(df["Age"].median()) # in sorted array by given value, position in the middle is even, if odd mean of two items in the middle
print(df["Age"].mode()) # most repeated value, useful for categorical data
print(df["Age"].std()) # square root of variance
print(df["Age"].var()) # measure of spread of data points around their mean (average) value

24.312599908574136
22.77799
0 18.0
Name: Age, dtype: float64
6.345968273732234
40.27131333121607
```

```
In [16]: print(df['MTRANS'].mode())
print(df['CALC'].mode())

0 Public_Transportation
Name: MTRANS, dtype: str
0 Sometimes
Name: CALC, dtype: str
```

```
In [17]: print(df['MTRANS'].unique())
print(df['CALC'].unique())

<StringArray>
['Public_Transportation', 'Walking', 'Automobile', 'Motorbike', 'Bike']
Length: 5, dtype: str
<StringArray>
['no', 'Sometimes', 'Frequently', 'Always']
Length: 4, dtype: str
```

```
In [18]: #column range - difference between the biggest and the lowest value

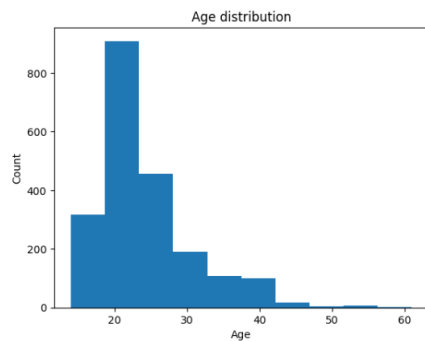
print(df["Weight"].max())
print(df["Weight"].min())

print(df["Weight"].max() - df["Weight"].min())

173.0
39.0
134.0
```

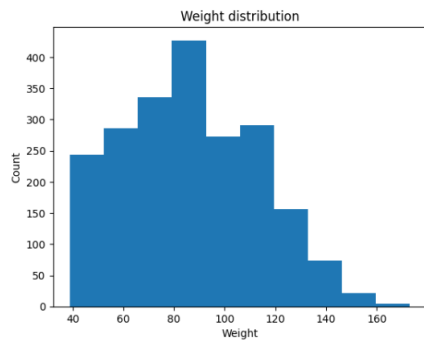
Visualization

```
In [19]: plt.hist(df['Age'])
plt.title("Age distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



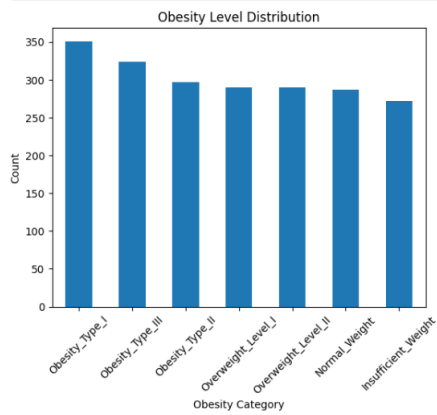
In [20]:

```
plt.hist(df['Weight'])
plt.title("Weight distribution")
plt.xlabel("Weight")
plt.ylabel("Count")
plt.show()
```



In [21]:

```
df['NObesyded'].value_counts().plot(kind='bar')
plt.title("Obesity Level Distribution")
plt.xlabel("Obesity Category")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.show()
```



In [22]:

```
sns.scatterplot(data=df, x='Height', y='Weight', hue='NObesyded')
plt.title("Height vs Weight by Obesity Level")
plt.show()
```

